

SEMICONDUCTORS

Q1. Mobilities of electrons and holes in a sample of intrinsic Ge at room temperature are $3600 \text{ cm}^2/\text{V-sec.}$ and $1700 \text{ cm}^2/\text{V-sec.}$ respectively. If the electron and hole densities are each equal to $2.5 \times 10^{19}/\text{m}^3$, calculate the conductivity.

Given: - $\mu_e = 3600 \text{ cm}^2/\text{V-sec} = 0.36 \text{ m}^2/\text{V-sec}$
 $\mu_h = 1700 \text{ cm}^2/\text{V-sec} = 0.17 \text{ m}^2/\text{V-sec}$
 $n_i = 2.5 \times 10^{19}/\text{m}^3$

Formula: - $\sigma_{\text{int}} = n_i(\mu_e + \mu_h).e$

Solution: - $\sigma_{\text{int}} = n_i(\mu_e + \mu_h).e$
 $= 2.5 \times 10^{19}(0.37 + 0.17) \times 1.6 \times 10^{-19}$
 $= 2.16 \text{ mho/meter}$

Ans: - conductivity is 2.16 mho/meter .

Q2. Find the resistivity of intrinsic Ge at 300° K . Given, density of carrier is $2.5 \times 10^{19}/\text{m}^3$. Mobility of electrons is $0.39 \text{ m}^2/\text{V-sec.}$ and mobility of holes is $0.19 \text{ m}^2/\text{V-sec.}$
Charge of electron is $1.6 \times 10^{-19} \text{ C}$.

Given: - $\mu_e = 0.39 \text{ m}^2/\text{V-sec}$
 $\mu_h = 0.19 \text{ m}^2/\text{V-sec}$
 $n_i = 2.5 \times 10^{19}/\text{m}^3$

Formula: - $\sigma_{\text{int}} = n_i(\mu_e + \mu_h).e$

$$\rho = \frac{1}{\sigma}$$

Solution: - $\sigma_{\text{int}} = n_i(\mu_e + \mu_h).e$
 $= 2.5 \times 10^{19} \times 1.6 \times 10^{-19} \times (0.39 + 0.19)$
 $= 2.32 \text{ mho/m}$
 $\rho = \frac{1}{\sigma} = 0.431 \text{ ohm-m}$

Ans: - resistivity is 0.431 ohm-m

Q3. An impurity of 0.01 ppm (particle per million) is added to Si. The semiconductor has a resistivity of 0.25 ohm-m at 300° K. Calculate the hole concentration and its mobility. Also comment on the results. Given, atomic weight of Si 28.1 and density of Si $2.4 \times 10^3 \text{ kg/m}^3$.

Given : - impurity level=0.01ppm, $\rho = 0.25 \text{ ohm-m}$, $T=300^0 \text{ K}$

$M= 28.1$, $n=2.4 \times 10^3 \text{ kg/m}^3$.

Formula: - $\rho = \frac{1}{en_h\mu_h}$

Solution: - no of Si atoms/unit volume

$$\begin{aligned} &= \frac{\text{avogadros no} \times \text{density}}{\text{atomic wt}} \\ &= \frac{6.023 \times 10^{23} \times 2.4 \times 10^3}{28.1} = 5.144 \times 10^{25} \text{ atoms/m}^3 \end{aligned}$$

Here,

1ppm= 1 impurity atom/ 10^6 Si atoms

1.01 ppm= 1 impurity atom/ 10^8 Si atoms

$\therefore$ no of impurity atoms

$$= \frac{5.144 \times 10^{25}}{10^8} = 5.144 \times 10^{17} \text{ atoms/m}^3$$

Each impurity contributes one hole

$\therefore 5.144 \times 10^{17}$ impurity atoms/ m^3 introduces the hole concentration of

$n_h = 5.144 \times 10^{17} \text{ holes/m}^3$

mobility, $\mu_h = \frac{1}{\rho en_h}$

$$= \frac{1}{0.25 \times 1.6 \times 10^{-19} \times 5.144 \times 10^{17}} = 48.6003 \text{ m}^2/\text{volt-sec}$$

$$\mu_h = 48.6003 \text{ m}^2/\text{volt-sec}$$

Ans: - hole concentration is 5.144×10^{17} holes/m³ and

Mobility of holes is 48.6003 m²/volt-sec.

Comment: - with the addition of impurity atoms, the hole concentration increases though its mobility remains the same.

Q4. The resistivity of Cu is 1.72×10^{-8} ohm-m. Calculate the mobility of electrons in Cu. Given, the number of electrons per unit volume is 10.41×10^{28} /m³.

Given: - $\rho = 1.72 \times 10^{-8}$ ohm-m
 $n_e = 10.41 \times 10^{28}$ /m³.

Formula: - $\mu = \frac{\sigma}{n_e \cdot e}$ and $\sigma = \frac{1}{\rho}$

Solution: - $\sigma = \frac{1}{\rho} = \frac{1}{1.72 \times 10^{-8}}$ mho/m
 $= 0.58 \times 10^8$ mho/m

$$\mu = \frac{\sigma}{n_e \cdot e} = \frac{0.58 \times 10^8}{10.41 \times 10^{28} \times 1.6 \times 10^{-19}} = 3.482 \times 10^{-3} \text{ m}^2/\text{V-sec}$$

Ans: - hence, mobility is 3.482×10^{-3} m²/V-sec

Q5. What is the probability of an electron being thermally excited to the conduction band at Si at 27° C. the band gap is 1.12 eV.

Given: - $T = 27^\circ \text{C} = 300\text{K}$, $E_g = 1.12\text{eV}$

$$K = 1.38 \times 10^{-23} \text{ J/K} = \frac{1.38 \times 10^{-23}}{1.6 \times 10^{-19}} = 86.25 \times 10^{-6} \text{ eV/K}$$

Formula: - $f(E_c) = \frac{1}{1 + e^{(E_c - E_f)/kT}}$

Solution: - Si is an intrinsic semiconductor. hence

$$E_c - E_f = \frac{E_g}{2} = 0.56\text{eV}$$

$$f(E_c) = \frac{1}{1 + e^{\frac{0.56}{86.25 \times 10^{-6} \times 300}}} = 3.9882 \times 10^{-10}$$

Ans: - probability is 3.9882×10^{-10}

Q6. In a solid there is an energy level lying 0.012eV below Fermi level. What is the probability of this level not being occupied by electron at 27° C?

Given: -

$$E_F - E = 0.012 \text{ eV}, \quad T = 27^\circ \text{C} = 300 \text{ K}$$

$$K = 1.38 \times 10^{-23} \text{ J/K} = \frac{1.38 \times 10^{-23}}{1.6 \times 10^{-19}} = 86.25 \times 10^{-6} \text{ eV/K}$$

Formula: - $f(E_C) = \frac{1}{1 + e^{(E_C - E_F)/kT}}$

Solution: -

Total probability = 1

Probability of occupying an energy + probability of not occupying the energy state = 1

$f(E) + \text{probability of not occupying the energy state} = 1$

probability of not occupying the energy state = $1 - f(E)$

$$\text{here, } f(E_C) = \frac{1}{1 + e^{(E_C - E_F)/kT}} = \frac{1}{1 + e^{\frac{0.012}{86.25 \times 10^{-6} \times 300}}} = 0.386$$

$$\text{hence, } 1 - f(E) = 1 - 0.386 = 0.614$$

Ans : - probability of not occupying is 0.614 or 61.4%

Q7. Fermi level in K is 2.1 eV. What are the energies for which the probabilities of occupancy at 300K are 0.99 and 0.01.

Given: - $E_F = 2.1 \text{ eV}$, $f(E_1) = 0.99$, $f(E_2) = 0.01$, $T = 300 \text{ K}$

$$K = 1.38 \times 10^{-23} \frac{\text{J}}{\text{K}} = \frac{1.38 \times 10^{-23}}{1.6 \times 10^{-19}} = 86.25 \times 10^{-6} \text{ eV/K}$$

Formula: - $f(E_C) = \frac{1}{1 + e^{(E_C - E_F)/kT}}$

Solution: -

$$f(E_1) = \frac{1}{1 + e^{(E_1 - E_F)/kT}}$$

$$0.99 = \frac{1}{1 + e^{(E_1 - E_F)/kT}}$$

$$1 + e^{(E_1 - E_F)/kT} = 1.01$$

$$e^{(E_1 - E_F)/kT} = 0.01$$

$$E_1 - E_p = -0.1192\text{eV}$$

$$E_1 = 1.9808\text{eV}$$

$$\text{And, } f(E_2) = \frac{1}{1+e^{(E_2-E_p)/kT}}$$

$$1.01 = \frac{1}{1+e^{(E_2-E_p)/kT}}$$

$$E_2 - E_p = 0.1189\text{eV}$$

$$E_2 = 2.2189\text{eV}$$

Ans : - The energies are 1.9808 eV and 2.2189 eV respectively

Q8. In n type semiconductor, the Fermi level lies 0.4eV below the conduction band .If the concentration of the donor atom is doubled, find the new position of the level w.r.t. the conduction band.

Given: -

$$E_c - E_f = 0.4\text{eV},$$

$$n_d' = 2n_d$$

$$T=300\text{K}$$

Formula: - $n_d = N_c e^{(E_F - E_C)/kT}$

Solution: -

$$n_d' = N_c e^{(E_F' - E_C)/kT}$$

$$\frac{n_d'}{n_d} = e^{[(E_F' - E_C) - (E_F - E_C)]/kT}$$

$$E_C - E_F' = (E_C - E_F) - kT \ln 2$$

$$= 0.4 - 0.018$$

$$E_C - E_F' = 0.3821\text{eV}$$

Ans: - The new Fermi level will be 0.3821eV below conduction band. Hence, the Fermi level will be shifted towards the conduction band by an amount $(0.4 - 0.3821) \text{eV} = 0.0179\text{eV}$.

Q9.In a Hall effect set up, a n-type Ge sample with donor concentration $2.5 \times 10^{21} / m^3$ is used. If the magnetic field is $0.5 \text{ wb} / m^2$, the current density is $500 \text{ A} / m^3$ and the thickness of the sample is 4mm, find the Hall voltage.

Given: - $n_e = 2.5 \times 10^{21}$, $B = 0.5 \text{ wb} / m^2$, $J = 500 \text{ A} / m^3$
 $w = 4 \text{ mm} = 4 \times 10^{-3} m$

Formula: -

$$V_H = \frac{BJd}{n_e e}$$

Solution: -

$$V_H = \frac{0.5 \times 500 \times 4 \times 10^{-3}}{2.5 \times 10^{21} \times 1.6 \times 10^{-19}}$$

$$= 2.5 \text{ mV}$$

Ans: - The Hall voltage is 2.5mV.

Q10. Calculate the charge carrier density and electron mobility of n type Si specimen of which the conductivity and Hall coefficients are given as $950 \text{ mho} / m$ and $1.5 \times 10^{-4} m^3 / C$.

Given: -

$$\sigma = 950 \text{ mho} / m, \quad R_H = 1.5 \times 10^{-4} m^3 / C.$$

Formula: -

$$R_H = \frac{1}{n_e e}, \quad \mu_e = \sigma R_H$$

Solution: -

$$n_e = \frac{1}{R_H e} = \frac{1}{1.5 \times 10^{-4} \times 1.6 \times 10^{-19}} = 4.1667 \times 10^{18} / m^3$$

$$\mu_e = \sigma R_H = 950 \times 1.5 \times 10^{-4} = 0.1425 m^2 / V\text{-sec}$$

Ans: - The charge carrier density is $4.1667 \times 10^{18} / m^3$ and the

Electron mobility is $0.1425 m^2 / V\text{-sec}$

Q11. Determine the current density of a n type semiconductor sample with Hall electric field of 150V/m. If its Hall coefficient is $-0.0145 \text{ m}^3/\text{C}$ and the electron mobility is $0.39 \text{ m}^2/\text{V-sec}$, determine the Hall current density

Given: - $E_H = 150$, $R_H = 0.0145 \text{ m}^3/\text{C}$ $\mu_e = 0.39 \text{ m}^2/\text{V-sec}$

Formula: -

$$J = \sigma E_H, \quad \mu_e = \sigma R_H$$

Solutions: -

$$\sigma = \frac{\mu_e}{R_H} = \frac{0.39}{0.0145} = 26.8966 \text{ mho/m}$$

$$J = 26.8966 \times 150 = 4034.49 \text{ Amp/m}^2$$

Ans: - The current density is 4034.49 Amp/m^2

Q12. The mobility of holes is $0.025 \text{ m}^2/\text{V-sec}$. What would be resistivity of p type if the Hall coefficient of sample is $2.25 \times 10^{-5} \text{ m}^3/\text{C}$.

Given: - $\mu_h = 0.025 \text{ m}^2/\text{V-sec}$, $R_H = 2.25 \times 10^{-5} \text{ m}^3/\text{C}$.

Formula: -

$$\rho = \frac{1}{\sigma}, \quad \mu_h = \sigma R_H$$

Solution: -

$$\sigma = \frac{\mu_h}{R_H} = \frac{0.025}{2.25 \times 10^{-5}} = 1111.11 \text{ mho/m}$$

$$\rho = \frac{1}{1111.11} = 9 \times 10^{-4} \Omega - \text{m}$$

Ans: - The resistivity is $9 \times 10^{-4} \Omega - \text{m}$

Q13. Find the Hall coefficient of sodium assuming BCC structure for Na of lattice constant $4.28 \text{ \AA}$.

Given: - $a=4.28 \text{ \AA}$, Number of atoms/unit cell=2 for BCC

Formula: - $R_H = \frac{1}{n_e}$

Solutions :-

Volume of 1 unit cell = a^3

Number of atoms/unit volume = $2/a^3$

Assuming that each atom contributes one donor electron

$$n_e = \frac{2}{a^3} = \frac{2}{(4.28 \times 10^{-10})^3} = 2.5509 \times 10^{28} / m^3$$

$$R_H = \frac{1}{2.5509 \times 10^{28} \times 1.6 \times 10^{-19}} = 2.4501 \times 10^{-10} m^3 / C$$

Ans: - The Hall coefficient is $2.4501 \times 10^{-10} m^3 / C$